

8/7
Q) Define a structure in C and Cpp. This structure should hold the variables, compute the size of structure manually and display the size of function.

Q. Use size of function and display the size. Check both same or not. Justify or your

8/1

C program

#include <stdio.h>

struct Student

```
{
    int rollNo;
    char grade;
    float marks;
};
```

int main()

{

struct Student s;

int manualSize = 4+1+4;

printf("Manual Size = %d bytes\n",

manualSize);

printf("Size using sizeof() = %lu bytes\n",

sizeof(s));

if (manualSize == sizeof(s))

printf("Both sizes are SAME\n");

else

printf("Both sizes are DIFFERENT");

return 0;

}

o/p : Manual size = 9 bytes

Size using sizeof() = 12 bytes

Both sizes are DIFFERENT.

C++ program

```
#include <iostream>
using namespace std;
struct student
{
    int rollNo;
    char grade;
    float marks;
};
```

```
int main()
```

```
{
    student s;
    int manualSize = 4+1+4;
    cout << "Manual Size = " << manualSize << "bytes" << endl;
    cout << "Size using sizeof() = " << sizeof(s) << "bytes" << endl;
    if (manualSize == sizeof(s))
        cout << "Both sizes are SAME";
    else
        cout << "Both sizes are DIFFERENT";
    return 0;
}
```

O/P: Manual Size = 9 bytes

Size using sizeof() = 12 bytes

Both ~~are~~ sizes are DIFFERENT

Justification

Manual size = int(4) + char(1) + float(4) = 9 bytes

sizeof() usually gives 12 bytes

This is because the compiler inserts
empty bytes (padding) to ensure
properly in memory
padding makes the program access